Practice Set 12 Paper 2 – Urology

1. Q1. 65 yr male underwent RP for stage T1c Gleason's score (4+3). At 6 months post surgery, his S. PSA has risen to 2 ng/ml above his preop PSA value. Discuss treatment and prognosis
2. Q2. Outline management of muscle invasive TCC. give critical evaluation of various types of neobladder.
3. Q3a. Describe the principles of vascular anastomosis. Enumerate and describe options for vascular access surgery for HD
4. Q3b. Describe medical management of renal calculi
5. Q3c. Embryogenesis of urinary bladder
6. Q3d. Tamsulosin
7. Q3e. Valve bladder.
8. Q3f. Access techniques in dialysis

Practice paper compiled from the question bank.

Q1. 65 yr male underwent RP for stage T1c Gleason's score (4+3). At 6 months post surgery, his S. PSA has risen to 2 ng/ml above his preop PSA value. Discuss treatment and prognosis

Type: LAQ (25 marks) (biochemical recurrence also discussed in 2019-Summer Paper 4, Q2)

Assumed meaning: 6 months after RP (radical prostatectomy) the PSA (prostate-specific antigen) never became undetectable and is now ≥2 ng/mL, i.e. PSA persistence with early rise, suggesting residual or systemic disease.

Introduction

- After RP, PSA (half-life about 3 days) should become undetectable (<0.1 ng/mL) within about 6 weeks.
- **PSA persistence:** failure to become undetectable after RP. **BCR (biochemical recurrence):** a rise after an undetectable nadir.
- **Gleason 4+3 = grade group 3** (unfavourable intermediate risk).

Definitions of BCR after RP

Body	Definition
AUA (American Urological Association)	PSA ≥0.2 ng/mL
NCCN (National Comprehensive Cancer Network)	Failure to reach undetectable, two consecutive rises, or PSA ≥0.1 ng/mL
EAU (European Association of Urology) risk groups	**Low risk:** PSADT (PSA doubling time) >1 year **and** grade group <4. **High risk:** PSADT ≤1 year **or** grade group 4–5

Why is the PSA high? (sources)

1. **Residual local disease:** positive margin (the only modifiable factor), residual benign tissue (low, stable PSA).
2. **Occult nodal metastases** (limited or no lymph node dissection).
3. **Distant micrometastases:** likeliest with persistent PSA and a fast rise.

Evaluation

- **Repeat PSA** to confirm; calculate PSADT.
- **Review pathology:** pT stage, margin (length >1 mm, Gleason 4 at margin), seminal vesicles, nodes. **Also know:** CAPRA-S (Cancer of the Prostate Risk Assessment–Surgery) nomogram; Decipher genomic classifier.
- **PSMA (prostate-specific membrane antigen) PET/CT (positron emission tomography/computed tomography):** best test; detection about half below 0.5 ng/mL and nearly 100% above 2 ng/mL. Choline or fluciclovine if PSMA is unavailable.
- **mpMRI (multiparametric MRI) pelvis:** best for local recurrence in the prostatic fossa.
- CT and bone scan are rarely positive below PSA 10 ng/mL.

Treatment

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Confirm PSA, PSADT, pathology → PSMA PET ± mpMRI
 ├ Negative or fossa only → salvage RT to bed + ADT
 ├ Pelvic nodes only      → bed + pelvic node RT + ADT
 ├ Oligometastatic (≤3)   → ADT ± metastasis-directed therapy
 └ Polymetastatic         → treat as mHSPC: ADT + ARPI ± docetaxel
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1. Salvage RT (radiotherapy): the only curative option

- Dose **≥64–66 Gy** to the bed (SAKK 09/10: 70 Gy gave no gain over 64 Gy); 40–45 Gy to pelvic nodes.
- Best at the lowest PSA: each 0.1 ng/mL costs about 2.6% recurrence-free survival.
- **RADICALS-RT, RAVES, GETUG-AFU 17 (ARTISTIC meta-analysis):** early salvage at PSA 0.1–0.2 equals adjuvant RT and spares ≥60% from RT. These apply to men who become undetectable, not to this patient.

2. Add ADT (androgen deprivation therapy) to salvage RT: essential here

Trial	Arms	Result
RTOG 9601 (Radiation Therapy Oncology Group)	RT ± 2 years bicalutamide 150 mg; PSA 0.2–4.0	12-year OS (overall survival) 76.3% vs 71.3%
GETUG-AFU 16	RT ± 6 months goserelin; PSA 0.2–2.0	10-year BCR-free 64% vs 49%; MFS (metastasis-free survival) better; OS unchanged
SPPORT (RTOG 0534)	Bed / bed + 4–6 months ADT / pelvis + ADT	5-year freedom from progression 70.9% / 81.3% / 87.4%
RADICALS-HD	None vs 6 vs 24 months ADT	24 months better MFS than 6 months

- At PSA ≥2 ng/mL with grade group 3, give whole-pelvis RT with **long-course (about 2 years) ADT**.

3. Systemic therapy

- **PSMA-positive metastases:** ADT + ARPI (androgen receptor pathway inhibitor: abiraterone, enzalutamide, apalutamide) ± docetaxel (LATITUDE, ARCHES, PEACE-1, ARASENS).
- **Oligometastases (≤3):** MDT (metastasis-directed therapy) by SBRT (stereotactic body RT) delays ADT (**STOMP:** ADT-free survival 21 vs 13 months; **ORIOLE:** progression 19% vs 61%).
- **High-risk BCR not fit for salvage: EMBARK** (PSADT ≤9 months, PSA ≥1 ng/mL): enzalutamide + leuprolide, MFS HR (hazard ratio) 0.42.
- **Also know:** castrate testosterone <50 ng/dL; intermittent ADT acceptable (PR.7); observation only for EAU low-risk BCR or short life expectancy; salvage lymph node dissection is investigational.

Prognosis

- **Untreated BCR (Pound):** metastasis in about one-third, median 8 years; death a median 5 years later. Predictors: **BCR <2 years, PSADT <10 months, Gleason 8–10.**
- **This patient is poor-risk:** persistent PSA, high level within 6 months, likely short PSADT. Persistent PSA carries a higher risk of metastasis than late BCR.
- **Salvage RT:** one-third to one-half BCR-free at 4–5 years; poorer above PSA 0.5–1 ng/mL. Cancer-specific survival gain mainly with PSADT <6 months treated within 2 years.
- Lifelong PSA follow-up.

Diagram to draw: the treatment flowchart above, with PSA kinetics curve after RP.

Key points to remember:

- EAU high-risk BCR: PSADT ≤1 year or grade group 4–5.
- PSMA PET first: almost 100% detection above PSA 2 ng/mL.
- Salvage RT ≥64 Gy + ADT: RTOG 9601 (OS), GETUG-AFU 16, SPPORT (add pelvis).
- Pound: BCR <2 years, PSADT <10 months, Gleason 8–10 predict death.

Q2. Outline management of muscle invasive TCC. give critical evaluation of various types of neobladder.

Type: LAQ (25 marks) (neobladder also answered in Practice Set 4 Paper 2, Q3e and Practice Set 3 Paper 4, Q2)

A. Management of MIBC (muscle-invasive bladder cancer; TCC = transitional cell carcinoma)

Staging

- Re-TURBT (transurethral resection of bladder tumour) with muscle; **MRI with VI-RADS** (Vesical Imaging-Reporting and Data System; ≥3 suggests invasion); CT (computed tomography) urogram, chest CT; bone scan if symptomatic.

1. Cystectomy pathway (standard for cT2–T4a N0 M0)

- **Cisplatin-based NAC (neoadjuvant chemotherapy):** ddMVAC (dose-dense methotrexate, vinblastine, doxorubicin, cisplatin) or GC (gemcitabine–cisplatin) ×4. **ABC meta-analysis:** 5% absolute 5-year survival gain. **SWOG 8710:** pT0 38% vs 15%. **VESPER:** ddMVAC better PFS (progression-free survival) than GC.
- **NIAGARA:** perioperative durvalumab + GC improved event-free survival and OS (overall survival).
- **Cisplatin-ineligible:** GFR (glomerular filtration rate) <60 mL/min, ECOG (Eastern Cooperative Oncology Group) ≥2, hearing loss, neuropathy, NYHA (New York Heart Association) III–IV failure → direct cystectomy.
- **RC (radical cystectomy) + standard bilateral PLND (pelvic lymph node dissection)** within 12 weeks of diagnosis. Men: cystoprostatectomy. Women: anterior exenteration; organ-sparing in selected cases. **SWOG S1011:** extended PLND gave no survival gain.
- **Adjuvant:** cisplatin chemotherapy if no NAC; **nivolumab** for pT3–4 or N+ (**CheckMate 274:** disease-free survival 20.8 vs 10.8 months).

2. Bladder preservation: TMT (trimodality therapy)

- Maximal TURBT + RT (radiotherapy) + radiosensitiser (**BC2001:** mitomycin C + 5-FU [fluorouracil]). Select cT2, solitary, **no hydronephrosis, no CIS (carcinoma in situ)**. Salvage cystectomy 10–30%.
- **Partial cystectomy:** solitary dome tumour, no CIS.

3. Metastatic disease

- **EV (enfortumab vedotin) + pembrolizumab** first line (EV-302: about doubled OS vs platinum). Alternatives: platinum chemotherapy then **avelumab maintenance** (JAVELIN Bladder 100); erdafitinib for FGFR (fibroblast growth factor receptor) alterations.

B. Urinary diversion options

- Ileal conduit (simplest, commonest), continent cutaneous pouch, or **orthotopic neobladder**. No diversion is proven superior for QoL (quality of life).

C. Neobladder: principles

1. **Reliable outlet:** intact rhabdosphincter, tumour-free urethra (frozen section).
2. **Low pressure:** complete **detubularisation** and cross-folding abolish peristalsis; a **sphere** gives most volume (Laplace).
3. **Capacity:** 44 cm of folded ileum holds <200 mL at first, **500–600 mL by 1 year**.

- **Contraindications:** positive urethral margin, stricture, eGFR (estimated GFR) <35–40 mL/min, liver failure, inability to self-catheterise. Age, obesity and nodes are not bars.

D. Critical evaluation of neobladder types

Type	Bowel; ureteric technique	Strengths	Weaknesses
Studer	40–44 cm ileum + isoperistaltic afferent limb; refluxing	Simple, no staples, takes short ureters; stricture <3%; good long-term function	Night incontinence; reflux on cystogram (harmless)
Hautmann W	About 70 cm ileum; refluxing since 1997	Largest capacity, early continence; stricture fell 9.5% → 1%	More retention, acidosis, B12 loss
T pouch (Stein–Skinner)	44 cm + tunnelled T-limb flap valve	Antireflux; accepts dilated ureters	Complex; more secondary procedures (USC [University of Southern California] RCT)
Abol-Enein serous tunnel	40 cm W, serosal troughs	Antireflux; stricture about 3.8%	Long ureters needed; no dilated ureters
Hemi-Kock	Intussuscepted nipple	Good early results	Late nipple stenosis, stones; abandoned
Camey II; Padua (VIP)	65 cm or 42 cm ileum; Le Duc tunnel	Early designs	Le Duc stricture 7–29%
Mainz III; right colon; sigmoid (Reddy)	Colon ± ileum	When ileum short or diseased	High-pressure contractions, poor compliance, more leakage

Critical points

- **Reflux vs obstruction:** Valsalva voiding pressurises the whole system equally, so reflux is benign; every antireflux valve can obstruct. **USC RCT** (484 patients): Studer = T pouch for renal function and infection; T pouch needed more secondary procedures. Hence **Studer and Hautmann** are the most popular.
- **Segment:** ileum is most compliant; colon and stomach give high pressures.
- **Capacity trade-off:** a larger pouch gives earlier continence but more retention and metabolic load.
- **Robotic intracorporeal neobladder:** comparable outcomes; long-term data awaited.

E. Outcomes and complications

- **Day continence 80–90%; night 20–50%.** Retention: men 4–10%, **women 20–30%**.
- **Late:** hyperchloraemic metabolic acidosis, B12 deficiency, mucus, stones, ureteroileal stricture, pouch rupture, urethral recurrence (men about 7% at 5 years).
- **Follow-up:** blood gas, B12, upper-tract imaging, urethral cytology if CIS.

Diagram to draw: Studer pouch with afferent limb and urethral anastomosis.

> **Key points to remember:**
> - NAC cisplatin: 5% absolute survival gain; NIAGARA adds durvalumab.
> - Studer and Hautmann (refluxing) dominate: obstruction, not reflux, destroys kidneys.
> - USC RCT: Studer = T pouch; T pouch more reoperations.
> - Day continence 80–90%, night 20–50%; women retain 20–30%.

Q3a. Describe the principles of vascular anastomosis. Enumerate and describe options for vascular access surgery for HD

Type: SAQ (10 marks) (also answered in Practice Set 5 Paper 2, Q2)

Assumed meaning: HD = haemodialysis.

Principles of vascular anastomosis

Principle	Why
Proximal and distal control; systemic heparin, heparinised flush	Prevents thrombosis in the static column
Atraumatic handling (adventitia only); clamp clear of atheroma	Avoids intimal dissection
Healthy ends, no tension, twist or kink	Prevents stenosis and leak
Intima-to-intima eversion	Smooth endothelium faces the lumen
Wide opening: spatulate, or end-to-side elliptical arteriotomy	Avoids stricture
Monofilament polypropylene, double-armed (5-0/6-0 transplant; 6-0/7-0 wrist fistula)	Smooth, low drag, low thrombogenicity
Inside-out on the artery	Tacks down plaque and intima
Carrel triangulation or parachute; interrupted in children	Clear back wall; allows growth
Flush before last knot; leave "growth factor"; release venous clamp first	Removes air and clot; avoids purse-string

Options for HD vascular access

Option	Description
1. Non-tunnelled CVC (central venous catheter)	Dual-lumen 10–12 Fr in internal jugular or femoral vein; emergencies, days only
2. Tunnelled cuffed CVC	Right internal jugular; bridge or last resort; infection, central stenosis
3. Native AVF (arteriovenous fistula), **first choice**	Best patency, fewest infections; matures 6–8 weeks
a. Radiocephalic (Brescia–Cimino)	Wrist, most distal; preserves proximal sites
b. Brachiocephalic	Elbow; higher flow, more steal
c. Brachiobasilic transposition	Deep basilic vein moved superficially
4. AVG (arteriovenous graft)	PTFE (polytetrafluoroethylene) loop; usable in 2–3 weeks; more stenosis and thrombosis
5. HeRO graft (Haemodialysis Reliable Outflow)	Bypasses central vein occlusion into right atrium
6. Endovascular AVF	Percutaneous forearm anastomosis

- **Planning:** distal before proximal, non-dominant arm; Allen test, Doppler vein mapping; create AVF about 6 months before dialysis; preserve arm veins.
- **Complications:** non-maturation, stenosis, thrombosis, **steal** (banding, DRIL [distal revascularisation–interval ligation]), aneurysm, infection.

Diagram to draw: radiocephalic end-to-side fistula with triangulation sutures.

Key points to remember:

- Intima-to-intima, polypropylene, inside-out on the artery, flush before tying.
- AVF first, distal, non-dominant arm, 6 months ahead.
- Rule of 6s: flow >600 mL/min, >6 mm wide, <6 mm deep, 6 weeks.

Q3b. Describe medical management of renal calculi

Type: SAQ (10 marks) (also answered in 2020-Summer Paper 4, Q2)

Aims

- Relieve pain, dissolve suitable stones and **prevent recurrence** (up to 50% at 5 years).

1. Acute care

- **NSAIDs (non-steroidal anti-inflammatory drugs)** first line; opioids if refractory. Urgent drainage if infection with obstruction.
- **MET (medical expulsive therapy):** α-blocker (tamsulosin) for distal ureteric stones 5–10 mm after migration from the kidney; trial 4–6 weeks.

2. Dissolution (chemolysis)

- **Uric acid** (radiolucent): potassium citrate to alkalinise (EAU [European Association of Urology]: pH 7.0–7.2 for dissolution); allopurinol if hyperuricosuric.
- **Cystine:** alkalinise to pH 7.0–7.5 with thiol drugs. **Struvite:** acetohydroxamic acid occasionally dissolves.

3. Prevention: general measures (all formers)

- **Fluid for urine ≥2.5 L/day** (Borghi 1996: recurrence halved).
- **Normal calcium**, low salt, less animal protein, more fruit and vegetables (Borghi 2002).

4. Metabolic evaluation

- Stone analysis, serum calcium, uric acid, creatinine, PTH (parathyroid hormone) if needed; one or two **24-hour urines** in high-risk or recurrent formers.

5. Selective therapy

Abnormality	Drug and dose
Hypercalciuria	**Thiazide:** hydrochlorothiazide 25 mg twice daily, chlorthalidone 25–50 mg, indapamide; add potassium citrate (thiazide lowers potassium and citrate)
Hypocitraturia, distal RTA (renal tubular acidosis)	**Potassium citrate 30–60 mEq/day**: alkali load reduces tubular citrate reabsorption
Hyperuricosuria	Allopurinol 300 mg/day
Uric acid stones	Potassium citrate to **pH 6.0–6.5**; avoid >7.0 (calcium phosphate)
Enteric hyperoxaluria	Calcium with meals, low fat and oxalate, magnesium
Primary hyperoxaluria type 1	Pyridoxine; **lumasiran**
Cystinuria	Urine >3 L, pH 7.0–7.5; **tiopronin** or D-penicillamine; cystine <250 mg/L
Struvite	Complete clearance; AHA (acetohydroxamic acid) 250 mg two to three times daily (urease inhibitor)

- **Also know:** AUA (American Urological Association): thiazide or citrate for recurrent calcium formers even without abnormality; NOSTONE (2023): hydrochlorothiazide did not reduce recurrence; repeat 24-hour urine within 6 months of therapy, then yearly.

Diagram to draw: flowchart from stone analysis to selective therapy.

Key points to remember:

- Urine ≥2.5 L/day; normal calcium, low salt.
- Thiazide for hypercalciuria; potassium citrate 30–60 mEq/day for hypocitraturia.
- Uric acid pH 6.0–6.5; cystine pH 7.0–7.5 and <250 mg/L.

Q3c. Embryogenesis of urinary bladder

Type: SAQ (10 marks) (also answered in 2021-Summer Paper 3, Q1)

1. Division of the cloaca (weeks 4–6)

- The **urorectal septum** (Tourneux and Rathke folds) divides the endodermal cloaca into a dorsal **anorectal canal** and a ventral **UGS (urogenital sinus)**.
- **Mesonephric (Wolffian) duct entry** is the landmark:
  - **Cranial vesicourethral canal** (continuous with the allantois) → **bladder and pelvic urethra**.
  - **Caudal (definitive) UGS** → phallic urethra (male); distal vagina and vestibule (female).

2. Layers of the wall

- **Urothelium:** UGS endoderm.
- **Lamina propria, detrusor, serosa:** surrounding splanchnic mesenchyme; epithelium and mesenchyme induce each other.

3. Trigone (weeks 5–6)

- The **common excretory ducts** (mesonephric duct distal to the ureteric bud) dilate and are absorbed into the UGS.
- **Apoptosis** separates ureters from mesonephric ducts. Ureteric orifices move **cranially and laterally**; mesonephric ducts end caudally as ejaculatory ducts.
- **Classic view:** trigone muscle is ureteric. **Current view** (mouse lineage tracing): mostly bladder smooth muscle.
- **Weigert–Meyer rule:** bud too low → lateral orifice, short tunnel → VUR (vesicoureteral reflux); too high → caudal, medial, ectopic orifice.

4. Allantois and urachus

- **Months 4–5:** the bladder descends into the pelvis; the allantoic duct elongates and obliterates as the **urachus** → median umbilical ligament.

5. Function

- Urine production starts at **week 10**. Fetal bladder seen in about half at 10 weeks, almost all by 13 weeks; empties every 15–20 minutes. After **16 weeks** amniotic fluid is mainly fetal urine.

Anomalies explained by embryology

Anomaly	Basis
Exstrophy	Cloacal membrane not reinforced by infraumbilical mesoderm
Urachal (patent urachus, sinus, cyst, diverticulum)	Failed obliteration at different levels
Agenesis, duplication	UGS or trigone failure; septation
Congenital (Hutch) diverticulum	Detrusor deficiency at the ureteric hiatus
VUR, ectopic ureter	Abnormal bud position

Diagram to draw: cloaca divided by the urorectal septum, labelled vesicourethral canal, UGS, allantois.

Key points to remember:

- Urorectal septum weeks 4–6; mesonephric duct entry is the landmark.
- Trigone: common excretory duct absorption; mostly bladder muscle (current view).
- Urachus forms at months 4–5; urine from week 10; amniotic fluid mainly urine after 16 weeks.

Q3d. Tamsulosin

Type: SAQ (10 marks)

Introduction

- A **uroselective α1-adrenergic antagonist**, first-line drug for bothersome male LUTS (lower urinary tract symptoms) due to BPH (benign prostatic hyperplasia) (AUA [American Urological Association], EAU [European Association of Urology]).

Receptor basis

Subtype	Site	Effect of blockade
α1a (about 70% in prostate)	Prostate, bladder neck, prostatic urethra, iris dilator	Relieves dynamic obstruction; ejaculatory dysfunction; IFIS (intraoperative floppy iris syndrome)
α1b	Blood vessels	Hypotension (tamsulosin has 10-fold less affinity)
α1d	Detrusor, spinal afferents	Storage symptom relief

Mechanism

- Blocks noradrenaline at α1a receptors → relaxes prostatic and bladder-neck smooth muscle → lowers the **dynamic component** of BOO (bladder outlet obstruction). It does not shrink the prostate.

Pharmacology and dose

- **0.4 mg once daily** after the same meal (up to 0.8 mg); no titration.
- Modified release; hepatic metabolism (CYP3A4, CYP2D6); avoid strong CYP3A4 inhibitors (ketoconazole).
- Flow rises within 4–8 hours, symptoms by 1 week; 0.2 mg is not better than placebo.

Indications

1. **BPH/LUTS:** IPSS (International Prostate Symptom Score) falls about 30–40%; Qmax rises; storage and nocturia improve.
2. **Combination with 5ARI** (5α-reductase inhibitor) for prostate >30–40 mL: **CombAT** (dutasteride + tamsulosin) and MTOPS (combination cut progression by 66%).
3. **MET (medical expulsive therapy):** distal ureteric stones 5–10 mm; SUSPEND (2015) negative overall but included all sizes.
4. **AUR (acute urinary retention):** before trial without catheter.
5. **Also know:** ureteric stent symptoms; female functional outlet obstruction.

Adverse effects

- **Ejaculatory dysfunction** (dose-dependent; less than silodosin): retrograde ejaculation from bladder-neck relaxation, or anejaculation (emission failure).
- **IFIS:** floppy, billowing iris, iris prolapse, progressive miosis during cataract surgery; **highest risk with tamsulosin**. Stopping the drug does not remove the risk, so warn the ophthalmologist.
- Dizziness, rhinitis, headache; less orthostatic hypotension than non-selective agents. Caution: sulfonamide allergy.

Comparison

	Tamsulosin	Silodosin	Alfuzosin
Selectivity	α1a/α1d	Most α1a	Clinical only

	Tamsulosin	Silodosin	Alfuzosin
Ejaculatory effect	Moderate	Highest	Lowest

Diagram to draw: prostate cross-section showing α1a receptors in stroma and bladder neck.

Key points to remember:

- 0.4 mg once daily; α1a/α1d selective; no titration.
- IFIS highest with tamsulosin; persists after stopping.
- MET benefit in distal stones 5–10 mm.

Q3e. Valve bladder.

Type: SAQ (10 marks) (also answered in 2020-Summer Paper 2, Q3e)

Definition

- **Valve bladder syndrome** (Mitchell, 1982): upper tracts and renal function worsen after successful PUV (posterior urethral valve) ablation, without residual obstruction, because of bladder dysfunction.

Pathophysiology (vicious cycle)

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Detrusor hypertrophy, collagen, low compliance → poor sensation, rising residual
+ polyuria (nephrogenic diabetes insipidus) → bladder never empty
→ pressure transmitted to dilated upper tracts → they refill the bladder → renal damage
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- Three processes: **polyuria, poor sensation, residual urine.**
- Age pattern: hyperreflexia in infancy → better compliance in childhood → large atonic bladder (myogenic failure) after puberty.

Clinical features

- Late continence (about 5.5 years), incontinence, UTI (urinary tract infection), worsening hydronephrosis, rising creatinine. **New incontinence after being dry is a red flag.**

Evaluation

- Uroflowmetry and PVR (post-void residual) at each visit; ultrasound; creatinine; voiding diary.
- **Videourodynamics** when the picture changes and before transplantation.

Management (escalating ladder)

1. **Behavioural:** adequate fluids, timed and double voiding, biofeedback.
2. **Drugs:** anticholinergic (oxybutynin) for a small high-pressure bladder; α-blocker for high residual. Stop oxybutynin if residual rises.
3. **CIC (clean intermittent catheterisation)** if residual rises despite drugs.
4. **Overnight bladder drainage** for worsening hydronephrosis, UTI or renal function; it gives bladder and kidneys a daily rest.
5. **Mitrofanoff appendicovesicostomy** if urethral CIC is difficult (sensate urethra, high bladder neck).
6. **Augmentation** (rare) for a small, high-pressure bladder despite the above: **ureterocystoplasty** with the dilated ureter is preferred, avoiding bowel (mucus, acidosis, stones).

- Reflux is treated through the bladder; reimplantation is rarely needed. Manage CKD (chronic kidney disease).

Transplantation

- ESRD (end-stage renal disease) in 20–50%. Optimise the bladder first (videourodynamics, CIC or overnight drainage); graft survival is then similar.

Diagram to draw: vicious-cycle diagram of the valve bladder.

Key points to remember:

- Polyuria, poor sensation, residual urine.
- Overnight drainage breaks the cycle.
- Ureteric augmentation rather than bowel; optimise before transplant.

Q3f. Access techniques in dialysis

Type: SAQ (10 marks) (CAPD also answered in Practice Set 1 Paper 2, Q3a)

Assumed meaning: access for PD (peritoneal dialysis) and HD (haemodialysis); emphasis on PD catheter insertion, since HD vascular access is answered in full in Practice Set 12 Paper 2, Q3a.

A. Peritoneal dialysis access

Catheter: Tenckhoff silicone catheter with **two Dacron cuffs**. The deep cuff sits in the rectus muscle; the superficial cuff about 2–4 cm from the exit. The tip lies in the **pelvis (rectovesical or rectouterine pouch)**; the exit site points **downwards** to reduce infection.

Insertion techniques

Technique	Steps	Comment
Open (mini-laparotomy)	Paramedian incision → split rectus → open posterior sheath and peritoneum → catheter to pelvis on a stylet → purse-string round deep cuff → tunnel → exit site	Simple; general or local anaesthesia
Laparoscopic (advanced)	Rectus sheath tunnel; **omentopexy**, adhesiolysis, hernia repair as needed	Best function; fixes causes of outflow failure
Percutaneous (Seldinger)	Needle, guidewire, dilator and peel-away sheath under local anaesthesia, ± ultrasound or fluoroscopy	Bedside; avoid after major abdominal surgery
Peritoneoscopic	Small-scope guided placement	Variant of percutaneous

- **Break-in:** delay full-volume exchanges about 2 weeks to prevent leak; urgent start uses small supine volumes.
- **Complications:** outflow failure (constipation, omental wrap, migration), leak, hernia, exit-site or tunnel infection, peritonitis, bowel perforation.
- **Contraindications:** peritoneal scarring, uncorrectable hernia, inability to perform exchanges.

B. Haemodialysis access (brief)

- **Non-tunnelled CVC (central venous catheter):** dual-lumen 10–12 Fr, internal jugular or femoral vein by **ultrasound-guided Seldinger**; for AKI (acute kidney injury); avoid subclavian (central stenosis).
- **Tunnelled cuffed CVC:** right internal jugular, tip at the cavoatrial junction under fluoroscopy; bridge to AVF.
- **AVF (arteriovenous fistula)** first choice, then **AVG (arteriovenous graft)**; HeRO and endovascular AVF in selected patients (see Practice Set 12 Paper 2, Q3a).

Diagram to draw: Tenckhoff catheter with two cuffs, tunnel, downward exit and pelvic tip.

> **Key points to remember:**
> - Tenckhoff: two cuffs, pelvic tip, downward exit.
> - Laparoscopic placement with omentopexy gives best function.
> - Avoid subclavian lines; AVF first for long-term HD.

Rapid Revision Sheet – Practice Set 12 Paper 2

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. 65 yr male underwent RP for stage T1c Gleason's score (4+3). At 6 months post surgery, his S. PSA has risen to 2 ng/ml above his preop PSA value. Discuss treatment and prognosis (LAQ)

- EAU high-risk BCR: PSADT ≤1 year or grade group 4–5.
- PSMA PET first: almost 100% detection above PSA 2 ng/mL.
- Salvage RT ≥64 Gy + ADT: RTOG 9601 (OS), GETUG-AFU 16, SPPORT (add pelvis).
- Pound: BCR <2 years, PSADT <10 months, Gleason 8–10 predict death.

Q2. Outline management of muscle invasive TCC. give critical evaluation of various types of neobladder. (LAQ)

- NAC cisplatin: 5% absolute survival gain; NIAGARA adds durvalumab.
- Studer and Hautmann (refluxing) dominate: obstruction, not reflux, destroys kidneys.
- USC RCT: Studer = T pouch; T pouch more reoperations.
- Day continence 80–90%, night 20–50%; women retain 20–30%.

Q3a. Describe the principles of vascular anastomosis. Enumerate and describe options for vascular access surgery for HD (SAQ)

- Intima-to-intima, polypropylene, inside-out on the artery, flush before tying.
- AVF first, distal, non-dominant arm, 6 months ahead.
- Rule of 6s: flow >600 mL/min, >6 mm wide, <6 mm deep, 6 weeks.

Q3b. Describe medical management of renal calculi (SAQ)

- Urine ≥2.5 L/day; normal calcium, low salt.
- Thiazide for hypercalciuria; potassium citrate 30–60 mEq/day for hypocitraturia.
- Uric acid pH 6.0–6.5; cystine pH 7.0–7.5 and <250 mg/L.

Q3c. Embryogenesis of urinary bladder (SAQ)

- Urorectal septum weeks 4–6; mesonephric duct entry is the landmark.
- Trigone: common excretory duct absorption; mostly bladder muscle (current view).
- Urachus forms at months 4–5; urine from week 10; amniotic fluid mainly urine after 16 weeks.

Q3d. Tamsulosin (SAQ)

- 0.4 mg once daily; α1a/α1d selective; no titration.
- IFIS highest with tamsulosin; persists after stopping.
- MET benefit in distal stones 5–10 mm.

Q3e. Valve bladder. (SAQ)

- Polyuria, poor sensation, residual urine.
- Overnight drainage breaks the cycle.
- Ureteric augmentation rather than bowel; optimise before transplant.

Q3f. Access techniques in dialysis (SAQ)

- Tenckhoff: two cuffs, pelvic tip, downward exit.
- Laparoscopic placement with omentopexy gives best function.
- Avoid subclavian lines; AVF first for long-term HD.